

## July 9, 2026 – Complete Live Telemetry & Bluetooth Integration

Today I completed the final stage of the telemetry system by integrating the **BNO055 IMU** and implementing **Bluetooth Low Energy (BLE)** communication. I began by soldering the header pins onto the IMU breakout board before wiring it to the ESP32 over the I<sup>2</sup>C interface. After configuring the required libraries and updating the firmware, the IMU successfully provided live pitch and roll measurements, replacing the final simulated telemetry values.

With the IMU integrated, the telemetry system now provides real-time monitoring of:

- Battery Voltage
- Motor Temperature
- Wheel RPM
- Vehicle Speed
- Pitch
- Roll

I also implemented Bluetooth Low Energy (BLE) communication, allowing the ESP32 to wirelessly transmit live telemetry data. Using a BLE debugging application, I successfully connected to the ESP32, verified the custom BLE service and characteristics, and confirmed that all telemetry data was being transmitted correctly in real time.

With this milestone complete, the embedded telemetry platform now consists entirely of live sensor data and wireless communication. The next phase of the project will focus on developing a custom mobile application to display telemetry data in a user-friendly interface. Once the application is complete, I will begin designing a custom KiCad PCB to replace the breadboard prototype and integrate the ESP32, sensors, and power distribution into a permanent electronics module.



